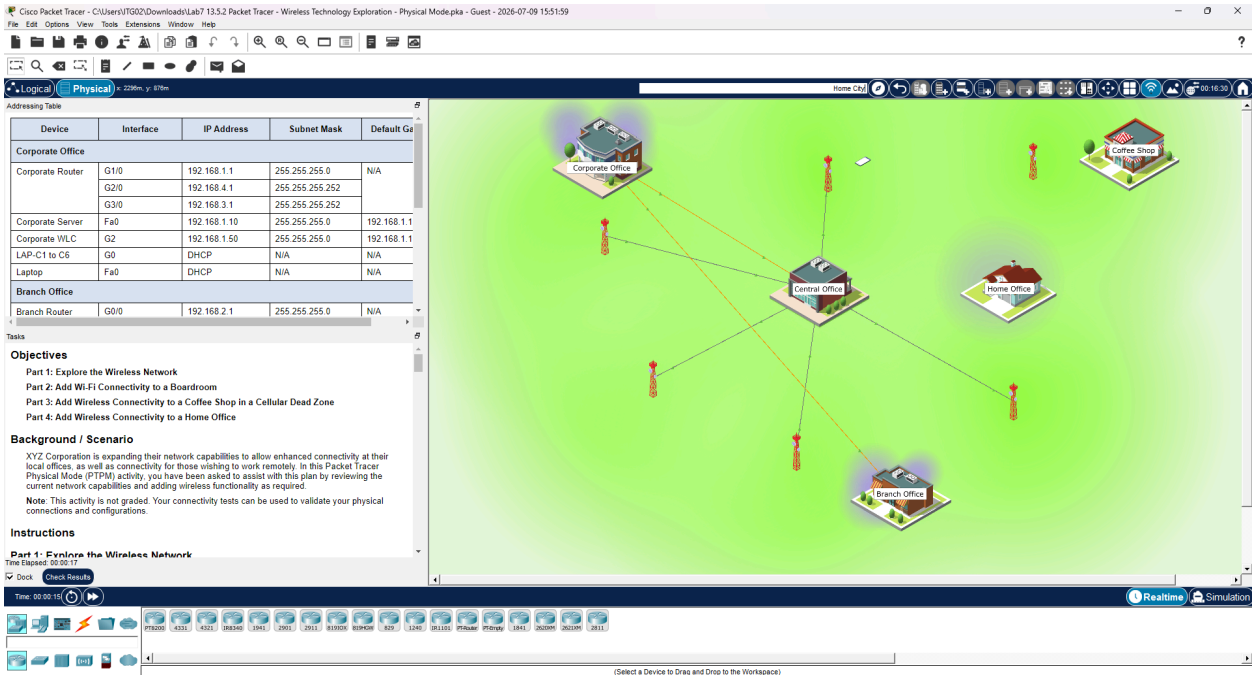
Lab Report: Wireless Network Expansion and Connectivity



1. Objectives

The primary goal of this activity was to expand the network capabilities of XYZ Corporation to support enhanced connectivity for local offices and remote workers. Key tasks included:

- Exploring the existing wireless topology and verifying end-to-end connectivity.
- Implementing Wi-Fi coverage in a boardroom "dead zone" within the Branch Office.
- Establishing cellular connectivity for a Coffee Shop located in a dead zone via tethering.

- Configuring a Home Office wireless router with WPA2-Personal security for remote access.

2. Topology Description

The network is spread across five primary locations within **Home City**:

1. **Corporate Office:** Houses the central server and a cluster of LWAPs managed by a WLC.
2. **Branch Office:** Contains a secondary server and localized LWAPs for office connectivity.
3. **Central Office (CO):** Serves as the backbone provider for cellular and internet services.
4. **Coffee Shop:** Initially a cellular dead zone, now connected via a tethered smartphone.
5. **Home Office:** A remote worker's residence equipped with a cable modem and wireless router.

3. Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Corpora...	G1/0	192.168.1.1	255.255.255.0	N/A
Corpora...	G2/0	192.168.4.1	255.255.255.252	N/A
Corpora...	G3/0	192.168.3.1	255.255.255.252	N/A

Device	Interface	IP Address	Subnet Mask	Default Gateway
Corpora...	Fa0	192.168.1.10	255.255.255.0	192.168.1.1
Corpora...	G2	192.168.1.50	255.255.255.0	192.168.1.1
Branch ...	G0/0	192.168.2.1	255.255.255.0	N/A
Branch ...	G2/0	192.168.4.2	255.255.255.252	N/A
Branch ...	Fa0	192.168.2.10	255.255.255.0	192.168.2.1
Branch ...	G2	192.168.2.50	255.255.255.0	192.168.2.1
CO Rou...	G0/0	192.168.5.1	255.255.255.252	N/A
CO Rou...	G1/0	192.168.6.1	255.255.255.252	N/A
CO Rou...	G3/0	192.168.3.2	255.255.255.252	N/A
CO Serv...	N/A	192.168.5.2	255.255.255.252	192.168.5.1
Cellular ...	N/A	172.16.1.1	255.255.255.0	N/A

4. Router & Wireless Configurations

4.1 Corporate & Branch Router Interface Logic (Cisco IOS)

The routers were configured to provide inter-office routing via point-to-point Gigabit links.

- **R1 (Corporate):** Configured G1/0 as the LAN gateway and G2/0 / G3/0 as backbone links.
- **Branch Router:** Configured G0/0 for the local branch network and G2/0 for the connection to the Corporate Office.

4.2 Home Wireless Router Settings

- **Internet Connection:** Automatic Configuration - DHCP.
- **LAN Setup:** IP 192.168.0.1, Subnet Mask 255.255.255.0.
- **DHCP Server:** Enabled (Range: 192.168.0.100 - .149).
- **Wireless SSID:** `homesweethome` (2.4 GHz).
- **Security:** WPA2-Personal, Passphrase: `mySecureNet`.

5. Lab Questions and Answers

Part 1: Explore the Wireless Network

Question: How are the four offices connected and what type of cables connect them?

Answer: The offices are connected via a combination of fiber optic and serial backbone links through the Central Office, establishing a Wide Area Network (WAN) architecture.

Question: How are the LWAPs connected to the network?
Answer: The Lightweight Access Points (LWAPs) are connected via Copper Straight-Through cables (Ethernet) to the local wiring closet switches.

Question: How are the cell towers connected to the Central Office Server?
Answer: The cell towers are connected using Coaxial cabling to the Central Office infrastructure.

Question: What are the different physical connections used between Smartphone1 and the Corporate Office Server?
Answer: The path includes: Wireless (Smartphone to Cell Tower) -> Coaxial (Cell Tower to CO Router) -> Fiber/Serial (CO Router to Corporate Router) -> Copper Ethernet (Corporate Router to Corporate Server).

6. Reflection

1. What overall benefit does wireless technology provide to the end user?
Wireless technology provides mobility and flexibility, allowing users to remain productive without being tethered to a physical desk. It also simplifies network deployment in areas where running physical cabling is difficult or costly.

2. Which form of wireless networking is the most beneficial?
This is subjective; however, Wi-Fi (802.11) is typically most beneficial for localized, high-bandwidth office tasks, while Cellular (4G/5G) is most beneficial for mobile workers requiring broad geographic coverage.

3. How could each of the following groups benefit from the wireless technologies presented?

- **Student:** Can access research materials and learning management systems from anywhere on campus or at home.

- **Salesperson:** Can maintain access to CRM data and client files while traveling or at client sites via cellular tethering.
- **R&D Engineer:** Can move freely with diagnostic equipment while maintaining a constant connection to laboratory servers.
- **Corporate Executive:** Gains the flexibility to conduct meetings and access corporate data across multiple office locations and during transit.